

Laboratory 5:

Signals and Systems

First-Order LTI Systems in Hardware

RC/RL Impulse and Step Responses

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Grading and Authenticity Requirements

Deliverable (1/2): This is a graded laboratory assignment. Include your **name and student ID** in all required screenshots and photographs.

Deliverable (2/2): All curve fitting and MATLAB verification must use waveform data acquired from your own physical circuit. Pre-computed synthetic traces are not accepted.

Checkpoint (1/6): Before submission, verify that each MATLAB figure includes axis labels with units, a legend, and your name/student ID in the title (or `sgtitle`).

1 Learning Objectives

By the end of this lab, you should be able to:

1. Derive and interpret first-order differential equations for RC and RL circuits.
2. Predict and measure time constants for step and impulse-like inputs.
3. Implement practical approximations of $u(t)$ and $\delta(t)$ using an Arduino digital output.
4. Acquire time-stamped ADC data and convert raw counts to volts in MATLAB.
5. Estimate empirical time constants $\hat{\tau}$ from measured data and quantify percent error versus theory.

2 Required Equipment and Software

Table 1. Required Hardware and Software

Category	Required Items
Controller	Arduino Uno (preferred) or Mega, USB cable
RC components	Bench-assigned resistor R , capacitor C
RL components	Bench-assigned resistor R_{ext} , inductor L with winding resistance R_L
Protection	Flyback diode (1N4148 or 1N4001), mandatory for RL circuit
Measurement	Bench multimeter (resistance and capacitance modes)
Prototyping	Breadboard and jumper wires
Software	Arduino IDE and MATLAB R2021a or later
Provided skeletons	<code>FastADC_Stream.ino</code> , <code>Lab05_Serial_Reader.m</code>

Arduino Pin Roles

Table 2. Pin Assignments

Signal	Pin	Direction	Purpose
Input source $x(t)$	D7	OUTPUT	Drives step/impulse-like input
Sense input	A0	INPUT	Measures circuit node voltage
Trigger indicator	D13	OUTPUT	LED toggles at acquisition start
Serial link	USB	TX/RX	115200 baud stream to MATLAB

3 Safety and Critical Wiring Notes

- Keep Arduino ADC input A0 within 0–5 V at all times.
- Never power the RL circuit before installing and verifying the flyback diode orientation.
- Disconnect USB power before rewiring between RC and RL circuits.
- Confirm a common ground between Arduino GND and breadboard ground rail.

Flyback diode orientation for RL setup:

- Diode anode $\rightarrow$ GND side of inductor.
- Diode cathode (striped end) $\rightarrow$ node between R_{ext} and L .

4 Background and Theory (Main Concepts)

4.1 What You Need in the Main Body

A first-order causal LTI system is governed by

$$\tau \frac{dy(t)}{dt} + y(t) = Kx(t). \quad (1)$$

The time constant τ sets the speed of response:

- At $t = \tau$, a rising step response reaches about 63.2% of its final value.
- At $t = \tau$, a decaying exponential falls to about 36.8% of its initial value.

4.2 RC Circuit Model

For the RC low-pass circuit with output across the capacitor,

$$RC \frac{dV_C}{dt} + V_C = x(t), \quad \tau_{RC} = RC. \quad (2)$$

For $x(t) = V_s u(t)$,

$$V_C(t) = V_s \left(1 - e^{-t/\tau_{RC}}\right). \quad (3)$$

4.3 RL Circuit Model Used in This Lab

The RL hardware wiring is series $R_{\text{ext}}-L$ to ground, with A0 sensing the junction node between R_{ext} and L . That node is the inductor-side voltage.

To match the analysis objective (response across R_{ext}), compute in MATLAB:

$$V_{R_{\text{ext}}}(t) = V_{\text{in}}(t) - V_{\text{node}}(t), \quad (4)$$

where $V_{\text{node}}(t)$ is measured by A0 and $V_{\text{in}}(t)$ is the known Arduino output level (approximately 5 V during the step window).

With inductor winding resistance R_L ,

$$R_{\text{eff}} = R_{\text{ext}} + R_L, \quad \tau_{RL} = \frac{L}{R_{\text{eff}}}, \quad K = \frac{R_{\text{ext}}}{R_{\text{eff}}}. \quad (5)$$

4.4 Impulse Approximation in Hardware

A finite pulse approximates an impulse only when $T_p \ll \tau$. In this lab, use TA-assigned T_p and verify $T_p/\tau \leq 0.1$ when possible.

4.5 Where Full Derivations Are

Full RC/RL derivations, detailed log-linearization equations, and extended mathematical notes are moved to Appendix A.

5 Pre-Lab Preparation

5.1 Bench Measurements (Complete at Start of Session)

Table 3. Measured Component Values

Symbol	Description	Measured Value
R	RC resistor	_____ Ω
C	RC capacitor	_____ μF
R_{ext}	RL external resistor	_____ Ω
L	Inductor	_____ mH
R_L	Inductor winding resistance	_____ Ω
R_{eff}	$R_{\text{ext}} + R_L$	_____ Ω
T_p	TA-assigned pulse width	_____ μs

5.2 Required Pre-Lab Work

Pre-lab Deliverable (1/1): Complete the following before running hardware:

1. Compute $\tau_{RC} = RC$ and $\tau_{RL} = L/(R_{\text{ext}} + R_L)$ using measured values.
2. Write analytical step-response expressions for RC and RL cases.
3. Compute expected impulse-approximation peak values using your TA-assigned T_p .
4. Generate a pre-lab MATLAB theory figure showing RC/RL step and impulse predictions.

Checkpoint (2/6): Show handwritten derivations and numerical time constants to the TA for CP-1 sign-off.

CP-1 Evidence: Handwritten derivations with name/student ID, including numerical τ_{RC} and τ_{RL} .

6 Lab Procedure (In-Lab Execution)

6.1 Part 1: Build and Verify RC Circuit

1. Wire $D7 \rightarrow R \rightarrow \text{node} \rightarrow C \rightarrow \text{GND}$.
2. Connect A0 to the node between R and C .
3. Connect Arduino GND to breadboard GND.
4. Before power-on, visually verify one complete current path and one common ground.

Checkpoint (3/6): Power on and trigger one acquisition. Confirm serial data values change over time (not constant).

RC Wiring Evidence: Photo with visible student ID and labeled RC node measured by A0.

6.2 Part 2: Build and Verify RL Circuit

1. Rewire as $D7 \rightarrow R_{\text{ext}} \rightarrow \text{node} \rightarrow L \rightarrow \text{GND}$.
2. Connect A0 to the same node between R_{ext} and L .
3. Install flyback diode: anode to inductor GND side, cathode to the node.
4. Verify diode orientation with stripe on cathode side before power-on.

Checkpoint (4/6): TA must verify diode orientation and node wiring before USB power is applied.

RL Wiring Evidence: Photo with diode orientation clearly visible and student ID present.

6.3 Part 3: Configure and Run Arduino Skeleton

Open `FastADC_Stream.ino`. Only edit configuration constants.

- Set `SAMPLE_RATE_HZ` so that $f_s \geq 10/\tau_{\min}$.
- Set `N_SAMPLES` to capture at least 5τ .
- Set `PULSE_WIDTH_US` to TA-assigned T_p .
- Run separate bursts for `STEP` and `IMPULSE`.

Checkpoint (5/6): In Serial Monitor (115200 baud), send `s` and confirm at least 10 lines of `timestamp_us,adc_count`.

CP-2 Evidence: Serial Monitor screenshot showing burst data after trigger.

6.4 Part 4: MATLAB Acquisition and Analysis Workflow

Use `Lab05_Serial_Reader.m` **skeleton**. Keep acquisition code unchanged; implement analysis blocks.

Task 1: Overlay Measured vs Theory

For each waveform (RC step, RC impulse, RL step, RL impulse):

1. Plot measured voltage versus time.
2. Overlay theoretical curve.
3. Add axis labels (units), legend, and name/student ID in title.

Important RL correction:

- A0 measures node voltage $V_{node}(t)$.
- Compute resistor response with $V_{R_{\text{ext}}}(t) = V_{in}(t) - V_{node}(t)$ before fitting RL step/impulse models.

Task 2: Extract $\hat{\tau}$ via Log-Linearization

1. Select post-transient fitting window.
2. Remove baseline offset.
3. Apply log transform to positive-valued transformed signal.
4. Use `polyfit` to get slope m and compute $\hat{\tau} = -1/m$.

Generate one diagnostic figure per waveform with:

- Top: original signal plus fitted exponential.

- Bottom: log-domain data plus regression line.

Task 3: Compute Percent Error

$$\text{Percent Error} = \left| \frac{\hat{\tau} - \tau_{\text{theory}}}{\tau_{\text{theory}}} \right| \times 100\%. \quad (6)$$

Report RC step, RC impulse, RL step, RL impulse in one summary table.

Checkpoint (6/6): Show TA: one overlay figure, one diagnostic fit figure, and your printed summary table for CP-3.

CP-3 Evidence: MATLAB window showing fitted $\hat{\tau}$ values and percent-error table.

7 Post-Lab Analysis and Deliverables

7.1 Required Discussion Questions

Answer each in 3–6 sentences with numerical evidence:

1. Compare RC and RL energy storage mechanisms and explain why RL DC gain includes R_L .
2. Evaluate impulse approximation quality using T_p/τ for both circuits.
3. Identify two non-ideal effects that cause log-domain curvature and describe mitigation.
4. Compute worst-case τ_{RC} bounds using tolerance assumptions and compare with measured $\hat{\tau}_{RC}$.
5. Numerically compare $h(t)$ and $ds(t)/dt$ from measured data and explain mismatch sources.

7.2 Submission Checklist

- ☐ Pre-lab theory figure (2x2) with name/student ID.
- ☐ Handwritten pre-lab derivations (RC and RL).
- ☐ RC and RL circuit photographs (ID visible).
- ☐ Serial data screenshot from CP-2.
- ☐ Four overlay plots (Task 1).
- ☐ Four log-linearization diagnostics (Task 2).
- ☐ Percent-error summary output (Task 3).
- ☐ Completed results table and TA sign-off page.
- ☐ MATLAB source code appendix.

7.3 Time-Constant Results Table

Table 4. Measured vs Theoretical Time Constants

Waveform	τ_{theory} (ms)	$\hat{\tau}_{\text{measured}}$ (ms)	Percent Error (%)
RC step response	_____	_____	_____
RC impulse response	_____	_____	_____
RL step response	_____	_____	_____
RL impulse response	_____	_____	_____

7.4 TA Checkpoint Sign-Off

Table 5. TA Checkpoint Sign-Off Sheet

CP	Requirement	Target Time	TA Initials
CP-1	Pre-lab derivations complete with measured component values and computed τ values.	0:00–0:15	_____
CP-2	RC and RL hardware verified; RL diode orientation approved before power.	0:15–1:05	_____
CP-3	MATLAB overlays, log-linear fit, and percent-error summary shown.	1:05–1:45	_____

Appendix A: Full Theory Derivations (Reference)

A.1 RC Derivation

Starting from KVL,

$$x(t) = V_R(t) + V_C(t) = RC \frac{dV_C}{dt} + V_C(t). \quad (7)$$

Hence,

$$RC \frac{dV_C}{dt} + V_C = x(t), \quad \tau_{RC} = RC. \quad (8)$$

For step input $x(t) = V_s u(t)$,

$$V_C(t) = V_s \left(1 - e^{-t/\tau_{RC}}\right) u(t). \quad (9)$$

For impulse response,

$$h_{RC}(t) = \frac{1}{\tau_{RC}} e^{-t/\tau_{RC}} u(t). \quad (10)$$

A.2 RL Derivation

With $R_{\text{eff}} = R_{\text{ext}} + R_L$ and current $i(t)$,

$$x(t) = L \frac{di}{dt} + R_{\text{eff}} i(t). \quad (11)$$

Output across the external resistor is

$$y(t) = V_{R_{\text{ext}}}(t) = R_{\text{ext}} i(t). \quad (12)$$

Then,

$$\frac{L}{R_{\text{eff}}} \frac{dy}{dt} + y(t) = \frac{R_{\text{ext}}}{R_{\text{eff}}} x(t), \quad (13)$$

so

$$\tau_{RL} = \frac{L}{R_{\text{eff}}}, \quad K = \frac{R_{\text{ext}}}{R_{\text{eff}}}. \quad (14)$$

For step input,

$$y(t) = K V_s \left(1 - e^{-t/\tau_{RL}}\right) u(t). \quad (15)$$

A.3 Log-Linearization

For decay $y(t) = A e^{-t/\tau} + C$,

$$\ln(y - C) = \ln A - \frac{t}{\tau}. \quad (16)$$

For rise $y(t) = Y_{\infty}(1 - e^{-t/\tau})$,

$$\ln(Y_{\infty} - y) = \ln Y_{\infty} - \frac{t}{\tau}. \quad (17)$$

In both cases, slope $m = -1/\tau$, so $\hat{\tau} = -1/m$.

Appendix B: Arduino Configuration and Example Logic

```
// Core user-editable constants in FastADC_Stream.ino
#define OUT_PIN      7
#define ADC_PIN      A0
#define SAMPLE_RATE_HZ 5000
#define N_SAMPLES    2000
#define PULSE_WIDTH_US 500
#define INPUT_TYPE    STEP    // STEP or IMPULSE
```

Set sample rate using $f_s \geq 10/\tau_{\min}$ and acquisition length for at least 5τ .

Appendix C: MATLAB Analysis Templates

C.1 RL Node-to-Resistor Conversion

```
% A0 measures node voltage V_node at junction of Rext and L.
% For RL analysis across Rext:
V_in = 5.0;           % during step-high window
V_Rext = V_in - V_node; % use this signal for RL curve fitting
```

C.2 Percent Error Template

```
pct_err = @(hat, theory) abs(hat - theory) / theory * 100;
```

Appendix D: Troubleshooting

- MATLAB hangs at `readline`: verify trigger character `s` is sent and sample counts match between Arduino and MATLAB.
- Flat trace at 0 V: A0 is not connected to the measurement node.
- Log plot is curved: baseline not removed, impulse too wide, or strong noise pickup.
- RL percent error much larger than RC: revisit R_L measurement and diode wiring; report non-ideal inductor effects in discussion.